





Walchand College of Engineering, Sangli.

(An Autonomous Institute)

Department of Information

Technology

A Project

Report On

Smart Crowd Monitoring and Choke Point Prevention System with Real-Time Density Visualization

Submitted in fulfillment for

B.Tech (IT)

Submitted by

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Academic Year: 2024-25

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Walchand College of Engineering, Sangli

(An Autonomous Institute)

Department Of Information Technology

CERTIFICATE

This is to certify that the Project Report entitled, **Smart Crowd Monitoring and Choke Point Prevention System with Real-Time Density Visualization** was submitted by

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to Walchand College of Engineering, Sangli, India, is a record of bonafide Project work carried out by them under my supervision and guidance and is worthy of consideration for the award of the degree of Bachelor of Technology in Information Technology of the Institute.

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Declaration

We, hereby declare that the project report entitled **"Smart Crowd Monitoring and Choke Point Prevention System with Real-Time Density Visualization"** submitted by us to **Walchand College of Engineering, Sangli** in fulfillment of the requirement for the award of the degree of **B. Tech** in **Information Technology** is a record of bonafide project work carried out by us under the guidance of **Dr D.B. Kulkarni.**

We further declare that the work reported in this project has not been submitted and will not be submitted, either in part or in full, for the award of any other degree or diploma in this institute or any other institute or university. We declare that this dissertation report reflects our thoughts about the subject in our own words. We have sufficiently cited and referenced the original sources, referred to or considered in this work. We have not misrepresented or fabricated or falsified any idea/data/fact/source in this submission. We understand that any violation of the above will be cause for disciplinary action by the Institute.

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Acknowledgement

We feel immense pleasure in submitting this Mini Project report entitled **"Smart Crowd Monitoring and Choke Point Prevention System with Real-Time Density Visualization"**

We are thankful to our guide **Dr D.B. Kulkarni** for their valuable guidance and kind help during the completion of our Project.

We are thankful to the Head of the Department **Dr. R. R. Rathod** for providing required infrastructure during the completion of the Mini Project I.

We would like to thank all faculty members and staff of the Department of Information Technology for their generous help in various ways to complete this thesis.

We would like to thank all our friends and especially our classmates for all the thoughtful and mind-stimulating discussions we had, which prompted us to think beyond the obvious. We have enjoyed their companionship so much during our stay at WCE, Sangli.

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1. Introduction

Smart Crowd Monitoring and Choke Point Prevention System with Real - Time Density Visualization is an intelligent, real-time system for monitoring and managing crowd behavior in sensitive areas. It uses computer vision, clustering, and fuzzy logic to detect crowd density, predict choke points, and suggest evacuation routes. Real-time heatmaps and path visualizations improve safety and situational awareness, making it a valuable tool for organizers and authorities.

1.1 Overview and Problem Statement

Managing large crowds in real-time is a complex task, especially in high-risk zones such as religious gatherings, political rallies, stadiums, or college festivals. Traditional surveillance systems often fail to provide predictive insights or intelligent responses to evolving crowd patterns. The proposed system aims to fill this gap by automatically detecting densely populated clusters using YOLOv8n for person detection and DBSCAN for clustering. It further employs Voronoi diagrams to identify nearby low-density zones and the concepts from graph theory to find possible choke points. The platform provides live visualization of density heatmaps and risk zones, enabling proactive measures to prevent stampedes and manage choke points efficiently.

1.2 Motivation

The motivation behind this project stems from several tragic incidents involving stampedes and chaotic crowd movement, which could have been avoided with timely monitoring and response. Manual monitoring is often slow, inconsistent, and reactive. Our goal is to create an intelligent, automated, and predictive system that can assist authorities in identifying critical crowd behavior, taking preventive action before a crisis escalates. By integrating real-time data visualization and risk analysis using fuzzy logic and graph-based techniques, this system ensures both public safety and efficient crowd flow management.

1.3 Contributions of this work

This project introduces a novel approach to real-time crowd analysis and management by combining computer vision, unsupervised clustering, and density visualization. Key contributions include:

1. Live detection of individuals using YOLOv8n to monitor crowd density in real-time and Weighted Moving Average (WMA) for predicting it.
2. Clustering of dense groups using DBSCAN to identify potential high-risk zones.
3. Generation of Voronoi diagrams to locate the nearest low-density areas.
4. Fuzzy logic risk scoring, graph isomorphism for symmetry detection, and heatmaps for density visualization.
5. A Tkinter-based GUI providing real-time updates, visual analytics, and alerts for possible stampede threats.

These innovations significantly enhance crowd safety, situational awareness, and decision-making in crowded environments.

2. Project Objectives

1. To design and develop a real-time monitoring system capable of detecting dense crowd clusters using YOLOv8n and DBSCAN, enabling early identification of potentially dangerous zones during large-scale events or campus activities.

2. To visualize density variations and potential escape routes using Voronoi diagrams. The goal is to assist security personnel and organizers in efficiently guiding crowds away from choke points.

3. To assess and classify risk levels using fuzzy logic and to implement graph-based techniques like isomorphism to detect symmetrical crowd formations that may become problematic.

4. To build a user-friendly and efficient interface using Python's Tkinter for live density visualization, path plotting, and alert notifications — creating a complete decision support system for crowd management.

3. Proposed Method / Algorithm

3.1 Problem Definition:

With rising re-identification risks from advanced analytics and external data sources, there is a need for a robust anonymization framework and safe data tool that ensures compliance with the Digital Personal Data Protection Act, 2023 - while maintaining an optimal balance between privacy protection and data utility.

3.2 Proposed Idea/ System:

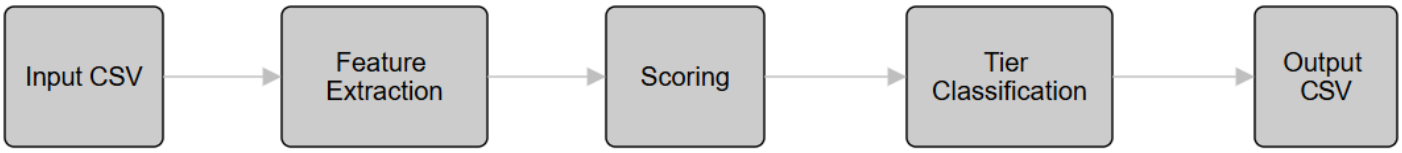
The Safe Data Anonymization and Utility Evaluation System is implemented in three key phases:

1. **Column Risk Tier Classification** – Identifies and categorizes dataset attributes based on their sensitivity and re-identification risk.
2. **Data Anonymization** – Applies suitable anonymization techniques (such as suppression, generalization, and PCA-based transformation) according to the risk tier.
3. **Utility Report Generation** – Evaluates the anonymized data to ensure an effective balance between privacy protection and data usability through visual and statistical reports.

PHASE 1: COLUMN RISK TIER CLASSIFICATION

Risk Distribution levels:

Tier 1 – Identifiers (High Risk)
Tier 2 – Quasi-Identifiers (Moderate Risk)
Tier 3 – Sensitive Attributes (Low Risk)



Pre-computation required before classification:

1. Brute-Force Keyword Matching

(i) Predefined list of common identifier column names (e.g., name, email,

aadhaar_no)
(ii) If a column matches, it is instantly classified as an Identifier — no computation needed.

2. Distinct Ratio (Uniqueness Measure)

$$\text{Distinct Ratio} = \frac{\text{Unique Values}}{n}$$

Higher ratio → more unique → higher chance of being an Identifier

3. Top-1 Frequency

$$\text{Top1 Frequency} = \frac{\text{Highest Frequency}}{n}$$

(i) Inversely related to identifiability.
(ii) Lower top-1 frequency → more likely an Identifier.

4. Shannon Entropy (Information Diversity)

$$\text{Normalized Entropy} = \frac{H(x)}{\log_2 d}$$

(i) Measures randomness of values
(ii) High entropy → more diverse → possible Identifier column.

5. Gini Impurity

$$corr(x, y) = \frac{cov(x, y)}{\sigma_x \sigma_y} = \frac{\Sigma(x - \overline{x})(y - \overline{y})}{\sigma_x \sigma_y}$$

(i) Indicates diversity (similar to entropy).
(ii) High impurity → diverse column values → potential identifier.

6. Skewness & Kurtosis (Outlier Detection)

(i) Applied to numeric columns.

(ii) Detects highly skewed or outlier-prone fields (e.g., unique IDs or codes).

7. Mutual Information (MI): Efficiently detects linkage-risk columns

Measures how strongly two attributes (random variables) are correlated.

$$I(X,Y) = \sum p(x,y) \log \left(\frac{p(x,y)}{p(x)p(y)} \right)$$

This formula is computationally expensive, especially for large datasets with many continuous variables.

Approximation:

$$MI_{approx}(X) = \frac{1}{n-1} \sum |\text{corr}(x_i, y_i)|$$

$$corr(x,\ y) = \frac{cov(x,\ y)}{\sigma_x \sigma_y} = \frac{\Sigma(x - \bar{x})(y - \bar{y})}{\sigma_x \sigma_y}$$

- Higher value → Column highly correlated with others → Likely Quasi-Identifier.
- Lower value→ Independent → Less risk of re-identification.

8. Scoring Mechanism

Each column receives a weighted composite score based on selected statistical features.

For Tier 1 (Identifiers)

$$S_{id} = 0.4(\text{Distinct Ratio}) + 0.3(\text{Entropy}) + 0.15(\text{Gini}) + 0.1(\text{Skewness}) + 0.05(\text{Kurtosis})$$

For Tier 2 (Quasi-Identifiers)

$$S_{qi} = 0.6(MI_{approx}) + 0.25(\text{Correlation Strength}) + 0.15(\text{Distinct Ratio})$$

For Tier 3 (Other)

$$S_{sens} = 0.5(\text{Entropy}) + 0.3(\text{Outlier Ratio}) + 0.2(\text{Distinct Ratio})$$

PHASE 2: DATA ANONYMIZATION

1. **k-Anonymity**
   (i) Ensures each record is indistinguishable from at least (k – 1) others.
   (ii) Reduces direct re-identification risk through generalization or suppression.
   For example: For all records r, $\exists$ k−1 others with same quasi-identifier values.

2. **l-Diversity**
   (i) Requires >= l well-represented distinct sensitive values in every equivalence class.
   (ii) Prevents Homogeneity and Background Knowledge Attacks that bypass k-Anonymity.

3. t-Closeness

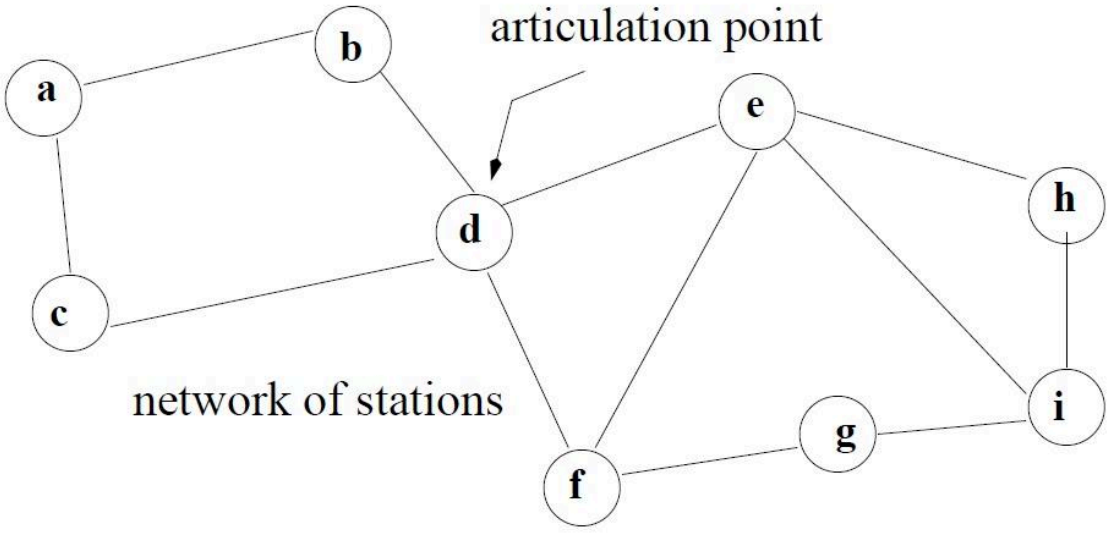
Ensures that the distribution of sensitive attributes within each equivalence class is close to the distribution in the entire dataset.

- Smaller t → Closer the local and global distributions → Higher privacy
- Larger t → More deviation allowed → More data utility

(ii) Articulation Points: Detecting Choke Points

A node (or person/zone) whose removal would **split the graph** into disconnected parts.

1. These nodes represent **critical zones or individuals** through which **most movement passes**.
2. If a crowd clusters near an articulation point, that area becomes a **choke point**—a failure or blockage there could trigger chaos.



(d) Z-test for Anomaly Detection

1. Monitors real-time crowd counts and compares them with historical average data.
2. Calculates the Z-score to measure how far the current count is from the mean.
3. If Z > 2, the crowd count is considered an anomaly (unusual surge).
4. Helps detect sudden spikes in specific areas before they become risky.
5. Triggers alerts for further clustering or evacuation analysis.

$$Z = \frac{X - \mu}{\sigma}$$

X: Current crowd count in the region
μ: Historical average crowd count
σ: Standard deviation of past data

(e) DBSCAN Clustering & Voronoi Diagrams for Empty Space Detection

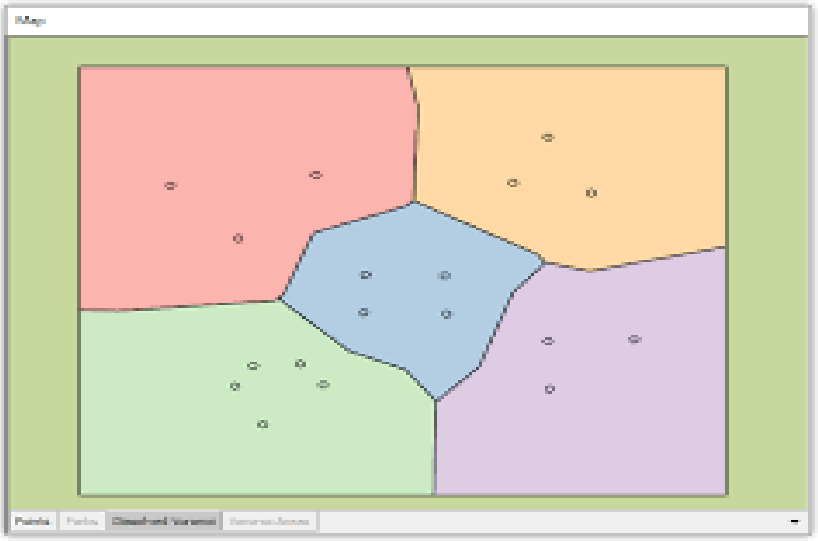
(i) Density-Based Spatial Clustering of Application with Noise

1. Groups nearby people into clusters based on distance and density.
2. Identifies **dense crowd areas** and marks isolated points as noise.
3. Helps locate **where people are concentrated**.

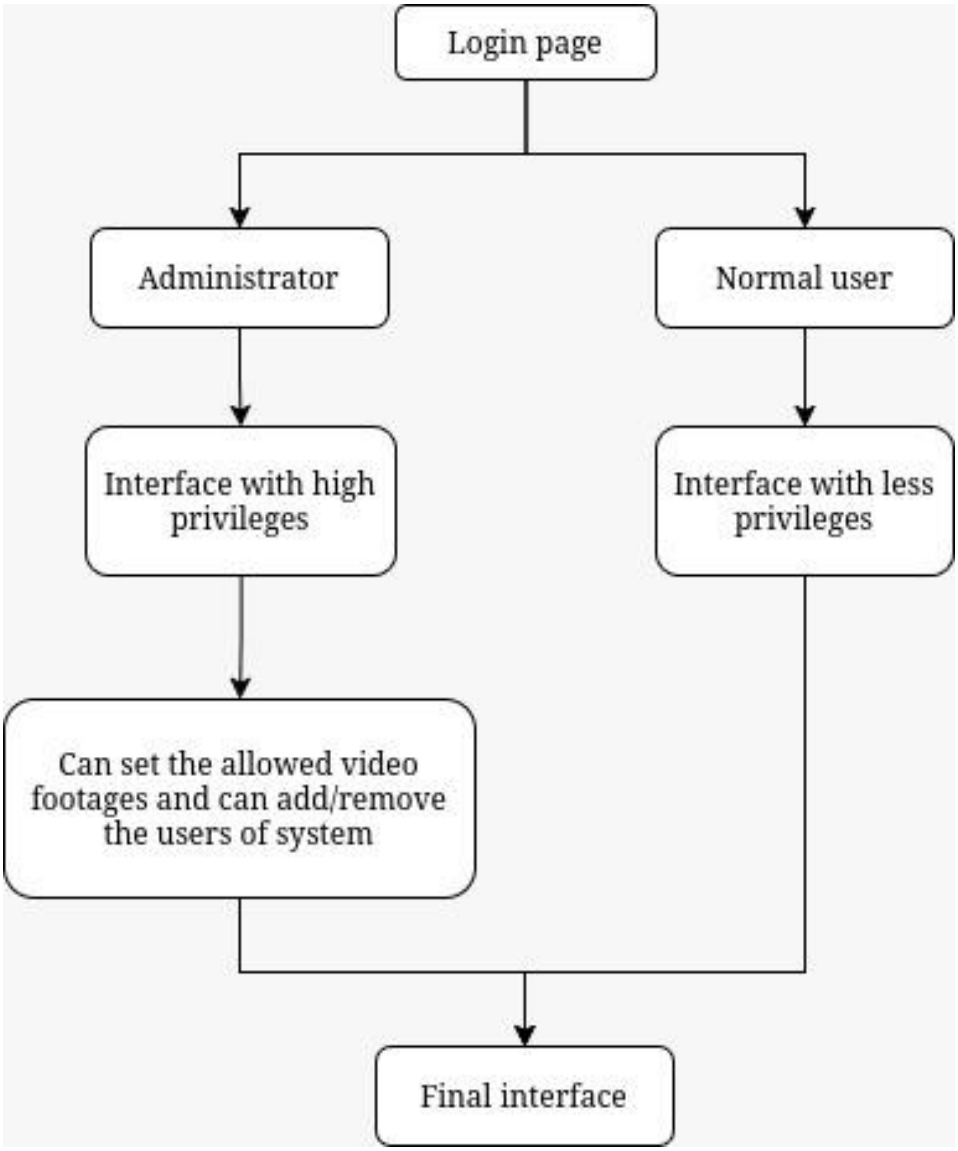


(ii) Voronoi Diagram

1. Divides the area into regions based on proximity to people/clusters.
2. **Larger Voronoi cells** represent **less crowded (empty) spaces**.
3. Helps **map low-density zones** that are safer or available for evacuation



3.3 System Analysis and Design:



4. Design and Implementation

4.1 Requirement specification:

For the Smart Crowd Monitoring and Choke Point Prevention System, the platform must prioritize **real-time performance**, **visual clarity**, and **crowd safety**, while maintaining ease of use for monitoring personnel. The system should meet the following specific requirements:

User Requirements:

1. Real-Time Crowd Visualization:
The platform should display live crowd density maps, heatmaps, and escape routes with minimal latency, allowing authorities to monitor and act swiftly.

2. Accurate Threat Detection:
The system must integrate clustering algorithms (like DBSCAN), fuzzy logic, and anomaly detection (Z-test) to identify choke points and potential stampede zones.

3. Secure and Reliable Data Handling:
All crowd movement data, analysis logs, and risk levels should be securely processed and stored to maintain system integrity and assist in later auditing or review.

4. Intelligent Alerting System:
Automated alerts should notify security personnel when risk thresholds are crossed (e.g., Z-score anomaly or high fuzzy risk), helping prevent dangerous build-ups.

5. Simple Operator Interface:
The UI should be intuitive and responsive, enabling users to monitor video feeds, view risk maps, and interpret crowd analytics without technical expertise.

4.2 Design and test steps:

We followed a modular and scalable approach in designing the system, ensuring reliable deployment, high-speed detection, and secure remote access.

1. Deployment Infrastructure:

- Hosted the **frontend** on an **AWS EC2 instance and backend on a server** with an **NVIDIA GPU** and **CUDA support** to accelerate YOLOv8 inference.
- Configured **NGINX** as both a **web server** and **reverse proxy**, managing requests between the Flask backend and the React/Tkinter frontend interface.
- Enabled secure remote access through **Cloudflare tunnels**, bypassing public IP exposure while maintaining encrypted traffic.

2. Backend and Detection Logic:

- Used **YOLOv8** for real-time **crowd detection** from live video streams, leveraging the GPU for fast frame processing.
- Implemented the backend using **Flask**, handling model inference, clustering, and real-time communication with the frontend.
- Deployed the Flask backend via **Gunicorn** behind NGINX for production-level performance and concurrency handling.

3. Crowd Analysis Algorithms:

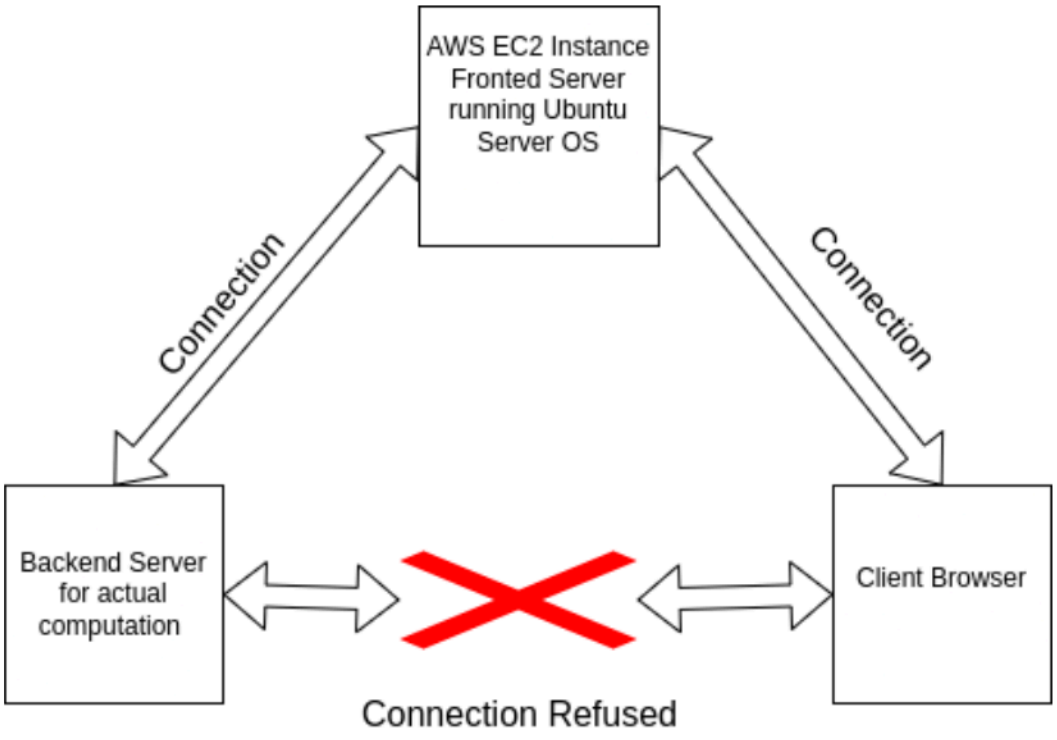
- Integrated **DBSCAN** to form clusters of densely packed individuals.
- Generated **Voronoi diagrams** to identify nearby low-density (safe) zones for crowd redirection.
- Used **Python-based modules** for calculating:
    - **Z-test scores** to flag anomalous crowd surges.
    - **Fuzzy logic risk scores** for dynamic risk classification.
    - **Graph theory techniques** (e.g., graph isomorphism & articulation points) to detect choke points and symmetrical threat patterns.

4. DNS & Domain Configuration:

- Integrated a **custom domain** with **DNS resolution** to provide easy access to the monitoring dashboard.
- Managed domain routing securely with **Cloudflare**, enabling HTTPS and access control.

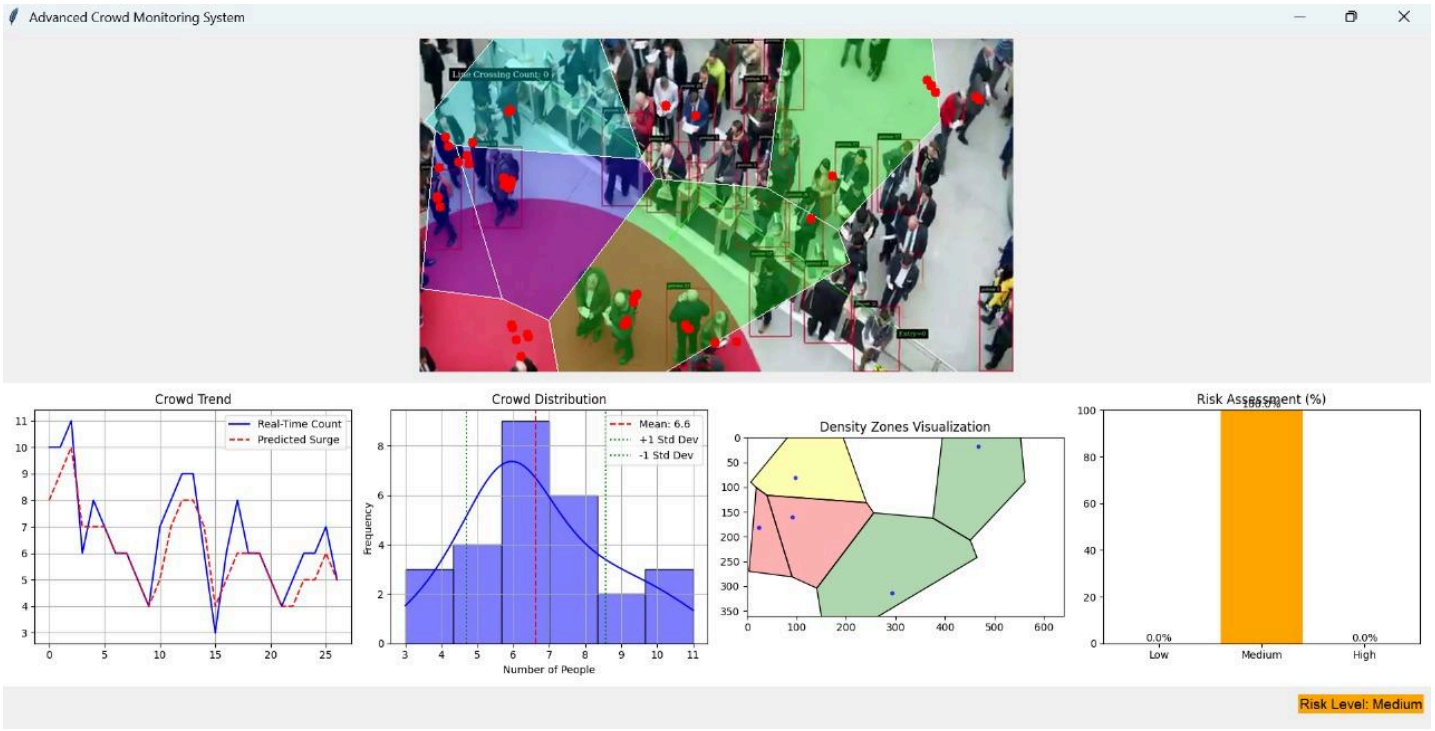
5. Testing and Optimization:

- Conducted **unit and integration testing** for every module—detection, clustering, risk analysis, and routing.
- Performed **load testing** on the EC2 instance to verify model performance under high crowd density.
- Tuned detection thresholds and clustering parameters for real-world scenarios to ensure system reliability and responsiveness.



5. Performance Study

5.1 Implementation/simulation environment



5.2 Summary of performance study:

The Smart Crowd Monitoring System shows strong performance in real-time detection, risk analysis, and crowd management. It accurately detects anomalies using Z-score, visualizes escape routes using A* or Dijkstra, and avoids obstacles like barricades during path planning. The system is optimized to reduce bandwidth usage and supports scalable, smooth operation even with high crowd density. Its secure cloud deployment ensures reliable access, making it an effective tool for preventing stampedes and managing crowd flow efficiently.

6. Conclusion

The Smart Crowd Monitoring and Choke Point Prevention System effectively combines advanced computer vision, clustering, fuzzy logic, and graph algorithms to enhance public safety. Its ability to detect anomalies, identify risky zones, and suggest real-time evacuation paths makes it a powerful tool for managing large crowds. With scalable performance, obstacle-aware routing, and bandwidth optimization, the system is well-suited for real-world deployment in high-density areas like festivals, stadiums, and public gatherings. It not only improves situational awareness for authorities but also plays a crucial role in preventing stampedes and ensuring safe crowd movement.

7. References

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